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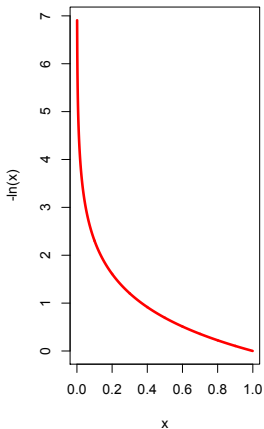
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Or more rigorously,

$$f_Y(y) = \begin{cases} f_X[g^{-1}(y)] \cdot \left| \frac{d}{dy} g^{-1}(y) \right| & \text{if } y = g(x) \text{ for some } x \\ 0 & \text{if } y \neq g(x) \text{ for all } x \end{cases}$$

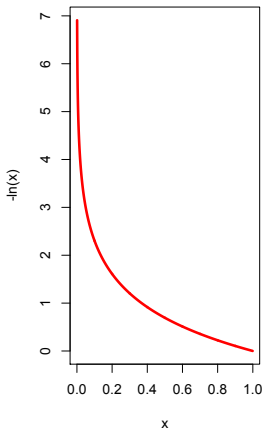
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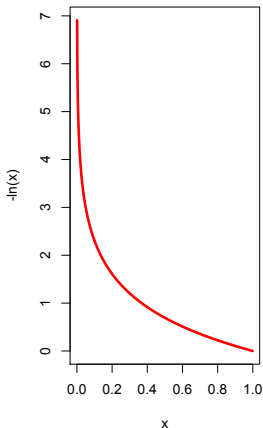
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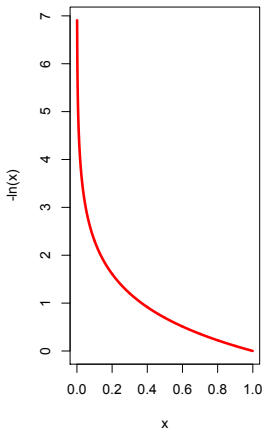
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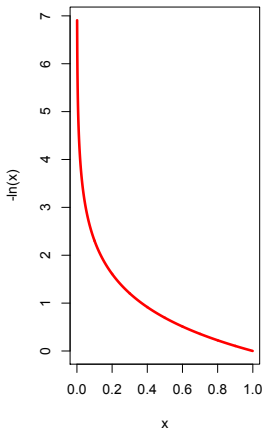
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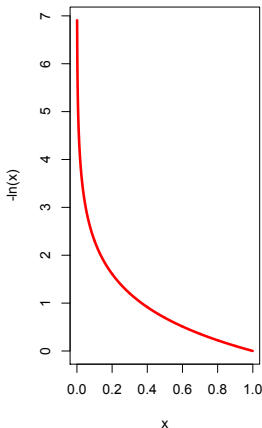
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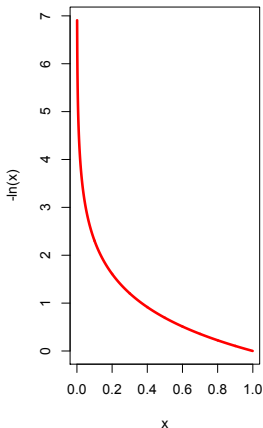
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$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| = 1 \cdot |-e^{-y}| = e^{-y}$$



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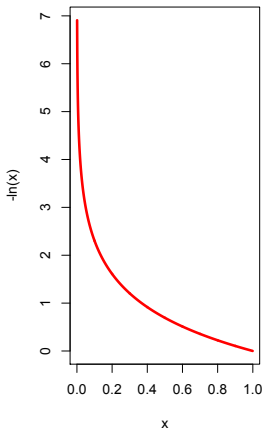
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$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right| = 1 \cdot |-e^{-y}| = e^{-y}$$

Therefore, Y has an exponential distribution,
 $Y \sim \text{Exp}(1)$



Let X have pdf $f_X(x)$, where $-\infty < x < \infty$. We want to find the pdf for $Y = |X|$. Can we use the formula $f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$?

- ☐ a) Yes
- ☐ b) No

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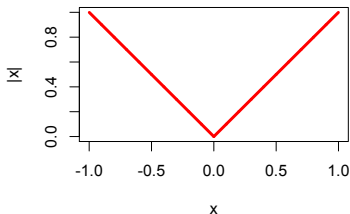
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(a) Yes

(b) No

$g(x) = |x|$ is not monotonic; also not differentiable at 0.

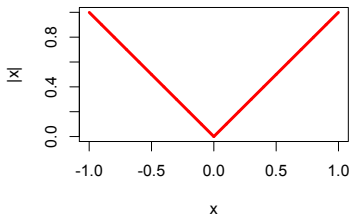


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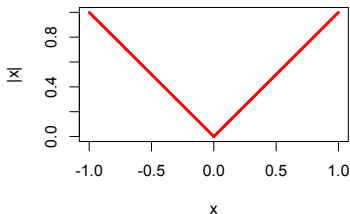
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First find cdf $F_Y(y)$, then find $f_Y(y)$.

For any $y \geq 0$,

$$\begin{aligned} F_Y(y) &= P(Y < y) \\ &= P(|X| < y) \\ &= P(-y < X < y) \\ &= F_X(y) - F_X(-y) \end{aligned}$$



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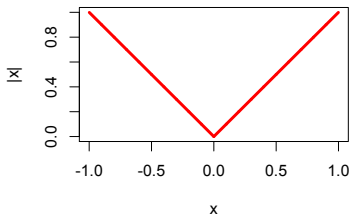
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$$\begin{aligned} f_Y(y) &= \frac{d}{dy} [F_X(y) - F_X(-y)] \\ &= f_X(y) + f_X(-y) \end{aligned}$$



Outline

- 1 Sum of two random variables

Sums of continuous random variables

If X, Y have a joint density $f(x, y)$, then $X + Y$ has the following density

$$\begin{aligned} f_{X+Y}(z) &= \int_{-\infty}^{\infty} f(x, z - x) \, dx \\ &= \int_{-\infty}^{\infty} f(z - y, y) \, dy \end{aligned}$$

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If X and Y are independent, we can use the convolution formula

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Cdf F_{X+Y} is called the convolution of the distributions F_X and F_Y .

Why?

$$F_{X+Y}(z) = \iint_{x+y \leq z} f_{X,Y}(x, y) \, dy \, dx$$

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$$\begin{aligned} F_{X+Y}(z) &= \iint_{x+y \leq z} f_{X,Y}(x,y) \, dy \, dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x,y) \, dy \, dx \\ f_{X+Y}(z) &= \frac{d}{dz} F_{X+Y}(z) \\ &= \int_{-\infty}^{\infty} \left\{ \frac{d}{dz} \int_{-\infty}^{z-x} f_{X,Y}(x,y) \, dy \right\} dx \\ &= \int_{-\infty}^{\infty} f_{X,Y}(x, z-x) \, dx \end{aligned}$$

Sum of two independent Uniforms:

Let X and Y have independent $\text{Unif}(0, 1)$ distribution. Find pdf of $X + Y$.

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For any $0 < z \leq 1$,

$$\begin{aligned} f_{X+Y}(z) &= \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx \\ &= \int_0^z 1 dx = z \end{aligned}$$

$f_{X+Y}(z)$

$$= \begin{cases} z & \text{if } 0 < z \leq 1 \\ 2 - z & \text{if } 1 < z < 2 \\ 0 & \text{otherwise} \end{cases}$$

For any $1 < z < 2$,

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Sum of two independent Uniforms: Triangular distribution

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Example: suppose $X, Y \stackrel{\text{ind}}{\sim} \text{Exp}(\lambda)$. What distribution does $X + Y$ have?

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For any $z > 0$,

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx \quad \text{Recall: Gamma distribution pdf is:}$$

$$= \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx \quad f_Z(z|\alpha, \lambda) = \frac{\lambda^\alpha}{\Gamma(\alpha)} z^{\alpha-1} e^{-\lambda z}$$

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Therefore, $X + Y \sim \text{Gamma}(2, \lambda)$.

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Sum of independent random variables

Random variables X and Y are independent, then

X	Y	$X + Y$
$N(\mu_1, \sigma_1^2)$	$N(\mu_2, \sigma_2^2)$	
$Poi(\lambda_1)$	$Poi(\lambda_2)$	
$Bin(n_1, p)$	$Bin(n_2, p)$	

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$Poi(\lambda_1)$	$Poi(\lambda_2)$	$Poi(\lambda_1 + \lambda_2)$
$Bin(n_1, p)$	$Bin(n_2, p)$	$Bin(n_1 + n_2, p)$